PHYS–467 Machine Learning for Physicists
Exercise session 5: Gradient Descent and some of its variants

Exercise n. 1

1. Load the MNIST dataset of hand-written digits and pre-process it;
2. Write a function that performs Gradient Descent (GD) with binary cross-entropy loss and l_2 regularization;
3. Plot the train loss and the validation loss as a function of the number of epochs;
4. Compute the generalization error.

Exercise n. 2

1. Write a function that performs Gradient Descent (GD) with binary cross-entropy loss and l_2 regularization plus early stopping;
2. Plot the train loss and the validation loss as a function of the number of epochs;
3. Compute the generalization error and compare it to the case without early stopping.

Exercise n. 3

1. Write a function that performs Stochastic Gradient Descent (SGD) (namely mini-batch GD with batch size = 1) with binary cross-entropy loss and l_2 regularization;
2. Plot the train losses of both GD and SGD as a function of the number of epochs;
3. Compute the generalization error and compare it to the case of GD.